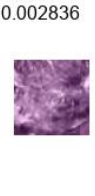
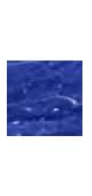
```
Arguments used:
 folder: C:\Users\pensf\Documents\AutomatedLBP_Benchmarking\LBP_Test_Images\Swatches\LBP_Texture_Swatches
 P: 8
 R: 1.0
 method: ror
 metric: chi2
 save_csv: matches.csv
 blur: 1.25
 ltp: True
 visualize: True
 X: 64
 Y: 64
 V: True
 G: False
Correct matches: 53/75 (70.67%)
Highest distance among correct matches: 0.038249
Lowest distance among correct matches: 0.001944
Highest distance among incorrect matches: 0.022133
Lowest distance among incorrect matches: 0.002836
```



```
Index: 0
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Chi Squared
Distance: 0.003429
```



```
Index: 1
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Chi Squared
Distance: 0.003676
```

```
Index: 2
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Chi Squared
Distance: 0.005913
```

```
Index: 3
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Chi Squared
Distance: 0.006000
```

```
Index: 4
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Chi Squared
Distance: 0.003676
```

```
Index: 5
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Chi Squared
Distance: 0.005584
```

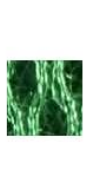
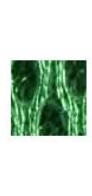
```
Index: 6
-- Original --
INSTANCE: brown
CATEGORY: brown
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: gray
-- Matched --
INSTANCE: gray
CATEGORY: gray
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: pink
Distance Metric: Chi Squared
Distance: 0.021531
```

```
Index: 7
-- Original --
INSTANCE: brown
CATEGORY: brown
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Chi Squared
Distance: 0.008813
```

```
Index: 8
-- Original --
INSTANCE: brown
CATEGORY: brown
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Chi Squared
Distance: 0.009532
```

```
Index: 9
-- Original --
INSTANCE: brown
CATEGORY: brown
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Chi Squared
Distance: 0.007898
```

```
Index: 10
-- Original --
INSTANCE: brown
CATEGORY: brown
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: brown
Distance Metric: Chi Squared
Distance: 0.006188
```

```
Index: 11
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Chi Squared
Distance: 0.004120
```

```
Index: 12
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Chi Squared
Distance: 0.004414
```

```
Index: 13
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Chi Squared
Distance: 0.004120
```

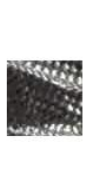
```
Index: 14
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Chi Squared
Distance: 0.003533
```

```
Index: 15
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Chi Squared
Distance: 0.003533
```

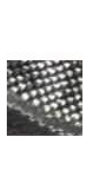
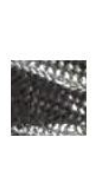
```
Index: 16
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Chi Squared
Distance: 0.006936
```

```
Index: 17
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: pink
Distance Metric: Chi Squared
Distance: 0.008986
```

```
Index: 18
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: pink
Distance Metric: Chi Squared
Distance: 0.014486
```

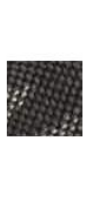


```
Index: 19
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Chi Squared
Distance: 0.012695
```

```
Index: 20
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: gray
-- Matched --
INSTANCE: gray
CATEGORY: gray
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Chi Squared
Distance: 0.038249
```

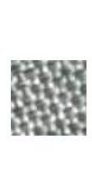
```
Index: 21
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: pink
Distance Metric: Chi Squared
Distance: 0.010259
```



```
Index: 22
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Chi Squared
Distance: 0.022133
```

```
Index: 23
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: green
-- Matched --
INSTANCE: green
CATEGORY: green
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: green
Distance Metric: Chi Squared
Distance: 0.010914
```

```
Index: 24
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: green
-- Matched --
INSTANCE: green
CATEGORY: green
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: green
Distance Metric: Chi Squared
Distance: 0.006116
```



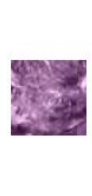
```
Index: 25
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: green
-- Matched --
INSTANCE: green
CATEGORY: green
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: green
Distance Metric: Chi Squared
Distance: 0.010914
```

```
Index: 26
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: green
-- Matched --
INSTANCE: green
CATEGORY: green
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: green
Distance Metric: Chi Squared
Distance: 0.006123
```

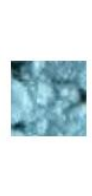
```
Index: 27
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: green
-- Matched --
INSTANCE: green
CATEGORY: green
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: green
Distance Metric: Chi Squared
Distance: 0.005904
```

```
Index: 28
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: green
-- Matched --
INSTANCE: green
CATEGORY: green
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: green
Distance Metric: Chi Squared
Distance: 0.005904
```

```
Index: 29
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Chi Squared
Distance: 0.002658
```



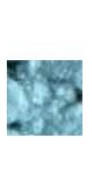
```
Index: 30
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Chi Squared
Distance: 0.003406
```



```
Index: 31
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Chi Squared
Distance: 0.002595
```

```
Index: 32
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Chi Squared
Distance: 0.004032
```

```
Index: 33
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Chi Squared
Distance: 0.003146
```



```
Index: 34
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Chi Squared
Distance: 0.002595
```

```
Index: 35
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Chi Squared
Distance: 0.003964
```

```
Index: 36
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: pink
Distance Metric: Chi Squared
Distance: 0.008499
```

```
Index: 37
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Chi Squared
Distance: 0.002836
```

```
Index: 38
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Chi Squared
Distance: 0.010689
```

```
Index: 39
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Chi Squared
Distance: 0.004024
```

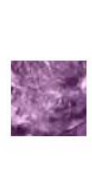
```
Index: 40
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: pink
Distance Metric: Chi Squared
Distance: 0.008499
```

```
Index: 41
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Chi Squared
Distance: 0.003168
```

```
Index: 42
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Chi Squared
Distance: 0.004478
```

```
Index: 43
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Chi Squared
Distance: 0.004802
```

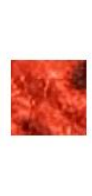
```
Index: 44
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Chi Squared
Distance: 0.004478
```



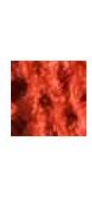
```
Index: 45
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Chi Squared
Distance: 0.003796
```

```
Index: 46
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Chi Squared
Distance: 0.003852
```

```
Index: 47
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Chi Squared
Distance: 0.001944
```



```
Index: 48
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Chi Squared
Distance: 0.003212
```



```
Index: 49
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Chi Squared
Distance: 0.001944
```

```
Index: 50
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Chi Squared
Distance: 0.003212
```

```
Index: 51
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Chi Squared
Distance: 0.002881
```

```
Index: 52
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Chi Squared
Distance: 0.002075
```

```
Index: 53
-- Original --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Chi Squared
Distance: 0.003904
```

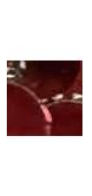
```
Index: 54
-- Original --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Chi Squared
Distance: 0.005022
```

```
Index: 55
-- Original --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Chi Squared
Distance: 0.003179
```

```
Index: 56
-- Original --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Chi Squared
Distance: 0.003179
```

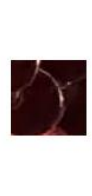
```
Index: 57
-- Original --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Chi Squared
Distance: 0.009376
```

```
Index: 58
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Chi Squared
Distance: 0.003505
```



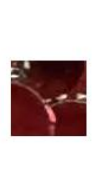
```
Index: 59
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Chi Squared
Distance: 0.006514
```

```
Index: 60
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: brown
-- Matched --
INSTANCE: brown
CATEGORY: brown
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Chi Squared
Distance: 0.006188
```

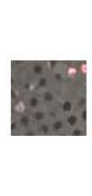


```
Index: 61
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Chi Squared
Distance: 0.003137
```

```
Index: 62
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Chi Squared
Distance: 0.003137
```

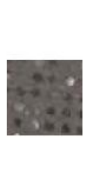


```
Index: 63
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Chi Squared
Distance: 0.006514
```

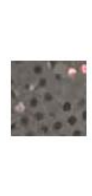


```
Index: 64
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Chi Squared
Distance: 0.004382
```

```
Index: 65
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Chi Squared
Distance: 0.004011
```



```
Index: 66
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Chi Squared
Distance: 0.008178
```



```
Index: 67
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Chi Squared
Distance: 0.004011
```

```
Index: 68
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Chi Squared
Distance: 0.004039
```

```
Index: 69
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Chi Squared
Distance: 0.006126
```

```
Index: 70
-- Original --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: yellow
-- Matched --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Chi Squared
Distance: 0.006561
```

```
Index: 71
-- Original --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: yellow
-- Matched --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Chi Squared
Distance: 0.003378
```

```
Index: 72
-- Original --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: yellow
-- Matched --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Chi Squared
Distance: 0.006561
```

```
Index: 73
-- Original --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: yellow
-- Matched --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Chi Squared
Distance: 0.003378
```

```
Index: 74
-- Original --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: yellow
-- Matched --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Chi Squared
Distance: 0.029632
```